Tedtronics — perfboard revision 5

7×10 cm stock · two banks · six interchangeable positions · three active capsules

ENGINEERING PROTOTYPE — NOT RELEASED FOR SOLDERING

Software checks: **pass**. Actual hardware, module envelopes, timing and electrical performance remain pending. Read the release gates before soldering.

Architecture

- ESP1 owns routing, rail enable and fault reset.
- ESP2 owns smart modes, voltage scanning and joystick ADC1 inputs.
- Each bank shares an ADS1115 voltage scanner and buffered selector.
- Every position retains independent output clamps and series-path isolation.
- Pi and its peripherals have a separate supply. RC522 uses the purchased MIFARE tags.
- Socket through-hole ICs. Carrier interfaces are labelled pigtails; their connector numbering is not IC numbering.

Construction comparison

Metric	Legacy	Revision 5
Boards	17	27
Functional parts	551	457
Adapter assemblies	47	23
Local wire segments	1511	1392
Board solder estimate	5224	4946

The legacy design used larger boards. Revision 5 also has 341 distribution posts and 279 harness conductors. Counts exclude carrier internals and adapter/socket preparation. This revision reduces adapters; it is not a claim of fewer boards or a finished minimum-work optimum.

Electrical bounds and release gates

Calculated design bounds

- Voltage input impedance: ≥1.98 MΩ with 1% divider resistors.
- Worst divided input at 5.25 V: 2.919 V.
- Static source/sink clamp maximum: 11.08 mA.
- Current shunt: 1 Ω ±1%; 10 mV shunt burden at 10 mA, plus switch and wiring resistance.
- Bank scan target: ≤20 ms. Wait for conversion-ready, discard after switching and keep timestamps.
- Watchdog expiry target: 200–350 ms; actual HC123 timing must be measured before permanent installation.
- 5 V guard static trip: 5.053–5.227 V. Approximately 23 mV margin below the contact ceiling does not establish safe fault transients.

These calculations do not establish loop stability, transient limits or measured accuracy.

Pending physical evidence

- Actual stock grid and plating
- Pogo pin diameter, compressed height and mounting fit
- DevKit pinout and body measurements
- Chosen ADS1115/INA226 carrier interfaces and envelopes
- Bench qualification of output clamps, supply guards and partial-power states
- Firmware implementation and measured acceptance

Verify the exact carrier pads, actual stock grid, socket bodies, compressed pogo heights and inter-board mounting offsets.

Current contact geometry needs **156 pogo pins**. The inventory lists 150, leaving a six-pin shortage before spares.

The 12 V adapter provides a nominal 24 W budget. Measure fan, light, controller and buck input loads. Do not include the Pi in this budget.

Board inventory

Board	Purpose	Parts	First phase
a1	A1 contacts	7	CP0
a2	A2 contacts	7	CP27
a3	A3 contacts	6	CP28
b1	B1 contacts	7	CP29
b2	B2 contacts	7	CP30
b3	B3 contacts	6	CP31
power-input	power input	13	CP9
power-guards	power guards	24	CP10
integration	Pi buttons / RC522 / enclosure harness	9	CP32
router-core	ESP1 carrier and enclosure I/O	20	CP0
router-safety	ESP1 watchdog / isolated returns	24	CP11
smart-core	ESP2 carrier and service pull-ups	27	CP15
smart-safety	ESP2 watchdog / relay registers	47	CP15
matrix-switches	Two MT8816 matrices / default reset	55	CP24
matrix-control	Matrix latch / local diagnostic buffer	45	CP24
power-analog	Student switch / current sense / comparator	27	CP11
power-interlock	Fault latch / watchdog interlock / test headers	19	CP9
measurement-a-precision	Bank A: buffered voltage scanner and continuity stimulus	38	CP1
measurement-a-buffers	Bank A: buffered voltage scanner and continuity stimulus	61	CP15
bank-a-position-0	Bank A: position 0 complete isolated channel	43	CP6
bank-a-position-1	Bank A: position 1 complete isolated channel	43	CP27
bank-a-position-2	Bank A: position 2 complete isolated channel	43	CP28
measurement-b-precision	Bank B: buffered voltage scanner	30	CP29
measurement-b-buffers	Bank B: buffered voltage scanner	61	CP29
bank-b-position-3	Bank B: position 3 complete isolated channel	43	CP29
bank-b-position-4	Bank B: position 4 complete isolated channel	43	CP30
bank-b-position-5	Bank B: position 5 complete isolated channel	43	CP31

CP00 — First light

Filmable win: ESP32 switches the real four-column LED loop.

Installation

Add 19 components/posts, 46 local wires and 0 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a1, router-core. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory LED: GPIO32 defaults LOW; explicit enable/disable; GPIO25 stimulus LOW; all future routing/output controls LOW; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Identify the exact classic ESP32 DevKit pinout, the USB-powered 5V output and 3V3 output. Identify actual pogo and adapter dimensions before soldering. Confirm the four kit pieces fit the selected rows.
• Unpowered A1: all six contacts within each column <1Ω; different columns >10MΩ with external links isolated. Inspect pin1 and the NODE header mapping.
• Prototype disconnected from A1: measure USB 5V ≤5.25V and 3V3 3.20–3.38V. R_LIMIT is 560Ω ±1%, ≥0.25W. With its output shorted, source current ≤9.5mA; remove short before connecting the kit.
• Place fake battery negative C1 / positive C2, 1kΩ resistor C2–C3, LED anode C3 / cathode C4, and kit jumper C4–C1. Prototype J_OUTPUT pin1→A1 J_NODE pin3 and pin2→pin1. No final-power or matrix cable connected.
• Compatible factory firmware: GPIO32 LOW/reset makes LED OFF; explicit enable lights LED; disable turns it OFF. Record USB-test-source voltage at C2 under load; it is intentionally unregulated and must not be called the final 5V rail.

Film the result

Show first light setup with power OFF and display firmware revision. Demonstrate: ESP32 switches the real four-column LED loop. Show the documented negative test and actual recorded readings.
Negative example: Disable or reset the ESP32: the LED must go dark without touching the kit.
Record actual readings, equipment, board/firmware revision and link positions.

CP01 — Reject an open contact

Filmable win: A lifted jumper contact is electronically rejected.

Installation

Add 38 components/posts, 66 local wires and 2 harness conductors. Earlier permanent work stays in place. Boards receiving parts: measurement-a-precision. • USB test-source positive lead unplugged; final supply and matrix leads absent.

• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.

• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.

• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Student supply OFF and positive fixed-battery lead unplugged. TEST_A→C4, TEST_B→C1; fit JP_RETURN only during test. Check stimulus current ≤0.34mA and educational contact voltages 0–5.25V.

• Calibrate open, a known short and 1kΩ. Measure open stimulus voltage Vs with DMM; infer R=11k×V/(Vs−V), accounting for divider gain and input loading. Known short ≤20Ω; ≥100Ω rejected; 20–100Ω or saturated/unstable readings inconclusive and OFF. Compare inferred short with DMM continuity <1Ω.

• Test the actual four-piece scene: correct C4–C1 kit jumper passes; one lifted contact and wrong-node placements fail. Include alternate paths through LED/resistor in calibration; no false pass through a component.

Film the result

Show reject an open contact setup with power OFF and display firmware revision.
Demonstrate: A lifted jumper contact is electronically rejected. Show the documented negative test and actual recorded readings.
Negative example: Lift the return jumper while retaining its convincing visual position: the electronic test rejects it.
Record actual readings, equipment, board/firmware revision and link positions.

CP02 — Reject a wrong node

Filmable win: A convincing-looking jumper on the wrong node is rejected.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB test-source positive lead unplugged; final supply and matrix leads absent.

• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.

• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.

• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Test the actual four-piece scene: correct C4–C1 kit jumper passes; one lifted contact and wrong-node placements fail. Include alternate paths through LED/ resistor in calibration; no false pass through a component.

• Simulated camera approval + failed/inconclusive electrical result keeps student supply OFF. Correction requires a fresh electrical test and separate enable request.

Film the result

Show reject a wrong node setup with power OFF and display firmware revision.

Demonstrate: A convincing-looking jumper on the wrong node is rejected. Show the documented negative test and actual recorded readings.

Negative example: Connect the jumper to a different column: electronic rejection persists despite simulated camera approval.

Record actual readings, equipment, board/firmware revision and link positions.

CP03 — Correction needs a fresh test

Filmable win: Correcting the jumper requires a fresh pass and a separate enable request.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB test-source positive lead unplugged; final supply and matrix leads absent.

• TEST_A→A1 C4, TEST_B→C1; JP_RETURN fitted only during diagnostic test.

• Before LED enable: stimulus OFF, JP_RETURN/probes removed, prototype positive restored with power OFF.

• Use bank-A precision input pair for TEST_A=C4 / TEST_B=C1 through removable sense leads. Its permanent NODE00/NODE01 harness is absent until CP06. Select CD4052 channel0; verify actual ADC/buffer supply before stimulus.

Firmware handoff

factory fixed-pair continuity: GPIO4 SDA,16 SCL,25 stimulus; ADS1115 0x48; manual source isolation; reject open/wrong/inconclusive, fresh test + explicit enable; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Before enable: stimulus OFF, JP_RETURN removed, test leads disconnected, fixed battery lead restored with power off. No automatic enable after a passing test.

• Simulated camera approval + failed/inconclusive electrical result keeps student supply OFF. Correction requires a fresh electrical test and separate enable request.

Film the result

Show correction needs a fresh test setup with power OFF and display firmware revision. Demonstrate: Correcting the jumper requires a fresh pass and a separate enable request. Show the documented negative test and actual recorded readings.

Negative example: Refit the correct jumper without a new test: it must remain OFF until a fresh pass and separate enable request.

Record actual readings, equipment, board/firmware revision and link positions.

CP04 — Capsule greeting

Filmable win: One capsule displays a real pt-BR greeting.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory OLED on ESP1: GPIO4 SDA /16 SCL, 0x3C; report identity and OFF/ready; no relay requests; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Connect only one OLED at 0x3C to ESP1 FIX_SDA GPIO4 / FIX_SCL GPIO16 using 3.3V logic and removable leads. Use one effective set of pull-ups; continuity fixture must be unplugged when it is not used. Confirm carrier pin labels before power.
• OLED displays greeting with educational source OFF. Display unplug/reconnect and reset do not automatically enable the source.

Film the result

Show capsule greeting setup with power OFF and display firmware revision.
Demonstrate: One capsule displays a real pt-BR greeting. Show the documented negative test and actual recorded readings.
Negative example: Unplug the OLED with power OFF: report missing hardware; a simulated greeting is not a pass.
Record actual readings, equipment, board/firmware revision and link positions.

CP05 — Identity and OFF status

Filmable win: The capsule displays its identity and actual OFF/ready state.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.

• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory OLED on ESP1: GPIO4 SDA /16 SCL, 0x3C; report identity and OFF/ready; no relay requests; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Display position A1, firmware revision, and Desligado/Pronto states. Never substitute a simulated GUI status for an actual hardware observation.

Film the result

Show identity and off status setup with power OFF and display firmware revision.

Demonstrate: The capsule displays its identity and actual OFF/ready state. Show the documented negative test and actual recorded readings.

Negative example: Change identity or remove the capsule: retain OFF and require a new configuration.

Record actual readings, equipment, board/firmware revision and link positions.

CP06 — Voltage calibration

Filmable win: The voltage input matches known zero and reference readings.

Installation

Add 14 components/posts, 25 local wires and 10 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a1, bank-a-position-0. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.

• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

• Reuse bank-A ADC/buffer. Remove diagnostic probes, JP_RETURN and stimulus leads; attach permanent NODE00/NODE01 harness. Connect service position0 after inspecting pull-ups; no duplicate ADC fixture.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Verify ADS1115 0x48. Sweep P1/P2 within 0–5.25V; compare displayed signed difference with DMM at 0,1,2.5,5V. Error ≤0.10V+5%|reading|, impedance ≥1MΩ, update rate ≥10Hz.

• Reverse sense polarity to show a negative difference; do not apply negative absolute contact voltages. P3 remains disconnected and ammeter paths open.

Film the result

Show voltage calibration setup with power OFF and display firmware revision.
Demonstrate: The voltage input matches known zero and reference readings. Show the documented negative test and actual recorded readings.
Negative example: Apply a known reference outside the accepted calibration tolerance: reject the calibration.
Record actual readings, equipment, board/firmware revision and link positions.

CP07 — Read real voltage

Filmable win: The capsule reading follows the multimeter as loop voltage changes.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• With prototype source and diagnostic probes isolated as specified, compare capsule V(P1)−V(P2) with DMM at zero and the actual limited-source voltage. Error ≤0.10V+5%|reading|; ≥10 updates/s. All node absolute voltages 0–5.25V.

Film the result

Show read real voltage setup with power OFF and display firmware revision.
Demonstrate: The capsule reading follows the multimeter as loop voltage changes.
Show the documented negative test and actual recorded readings.
Negative example: Remove the measured source: display the actual near-zero voltage, not a cached reading.
Record actual readings, equipment, board/firmware revision and link positions.

CP08 — Signed voltage

Filmable win: Swapping only the measurement leads reverses the reading sign.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • Prototype source only: positive C2 / ground C1; final supply and matrix disconnected.
• OLED uses ESP1 GPIO4 SDA / GPIO16 SCL at 3.3V. Remove diagnostic probes before ordinary operation.

Firmware handoff

factory voltage: ESP1 OLED + local ADS1115 0x48; USB3V3 to isolated card supply via removable links; P3/ammeter disconnected; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Power OFF before swapping P1/P2 removable measurement leads. Compare signed differential reading with DMM in both orientations; never apply a negative absolute node voltage. P3 and ammeter paths remain disconnected.

Film the result

Show signed voltage setup with power OFF and display firmware revision.
Demonstrate: Swapping only the measurement leads reverses the reading sign.
Show the documented negative test and actual recorded readings.
Negative example: Reverse the test leads: the sign must reverse with the same magnitude tolerance.
Record actual readings, equipment, board/firmware revision and link positions.

CP09 — Regulated rails

Filmable win: The isolated final regulators produce their specified voltages.

Installation

Add 17 components/posts, 26 local wires and 8 harness conductors. Earlier permanent work stays in place. Boards receiving parts: power-input, power-interlock. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Keep JP_LOGIC and JP_3V3 open while adjusting and qualifying both purchased buck outputs. Final consumers are not connected at this checkpoint.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Unpowered: every rail >1kΩ to GND after capacitor charging; no accidental node/service connection.

• With links open: 5V buck 4.90–5.04V; 3.3V buck 3.20–3.38V. Current-limited 12V supply, 100mA initially.

Film the result

Show regulated rails setup with power OFF and display firmware revision.

Demonstrate: The isolated final regulators produce their specified voltages. Show the documented negative test and actual recorded readings.

Negative example: Apply no input: outputs remain unpowered; never invent a regulator reading.

Record actual readings, equipment, board/firmware revision and link positions.

CP10 — Overvoltage guards

Filmable win: Each supply guard rejects an out-of-range source.

Installation

Add 24 components/posts, 44 local wires and 6 harness conductors. Earlier permanent work stays in place. Boards receiving parts: power-guards. • USB prototype source and prototype 3V3 links removed before final supplies connect. • Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Independent OV disconnects: sweep 5V/3V3 buck interfaces with consumers isolated; verify actual thresholds and strict contact peak ≤5.25V during a bounded 12V failed-buck step. Test both guards separately before connecting any IC.

Film the result

Show overvoltage guards setup with power OFF and display firmware revision.
Demonstrate: Each supply guard rejects an out-of-range source. Show the documented negative test and actual recorded readings.
Negative example: Raise each rail above its documented trip threshold using the bounded fixture: guard must reject.
Record actual readings, equipment, board/firmware revision and link positions.

CP11 — OFF at startup and reset

Filmable win: The final student rail remains OFF at startup and reset.

Installation

Add 53 components/posts, 112 local wires and 22 harness conductors. Earlier permanent work stays in place. Boards receiving parts: router-core, router-safety, power-analog, power-interlock. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Factory ALIVE fixture is temporary: J_PROTOTYPE pin2 USB3V3→J_POWER pin9 only during supervised load qualification. LOW/open for startup/reset; remove permanently before CP13.

• Remove the temporary USB-router-to-3V3_SYS feed before fitting qualified JP_LOGIC/JP_3V3 consumer links. Never parallel USB and buck rails.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Prototype USB positive and 3V3 supply links are removed before connecting final consumer links. Only one educational source may be connected. Inspect all power-board blocks before enabling; the student latch must not be powered as an incomplete circuit.

• Supervised factory qualification only: ALIVE is held LOW for startup/reset tests. All consumer links remain open until voltage guards pass. USB removal, held reset and a power cycle leave 5V_EDU OFF.

Film the result

Show off at startup and reset setup with power OFF and display firmware revision.
Demonstrate: The final student rail remains OFF at startup and reset. Show the documented negative test and actual recorded readings.
Negative example: Reset or reconnect USB with enable previously requested: startup remains OFF.
Record actual readings, equipment, board/firmware revision and link positions.

CP12 — Overload trip and latch

Filmable win: An overload shuts down and stays latched until a valid reset.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• Factory ALIVE fixture is temporary: J_PROTOTYPE pin2 USB3V3→J_POWER pin9 only during supervised load qualification. LOW/open for startup/reset; remove permanently before CP13.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• 1kΩ and 470Ω rail loads remain enabled; 330Ω or 100Ω load trips and remains latched. At 10mA: 5V_EDU 4.75–5.25V.
• Sweep load: trip threshold must be >10.6mA and ≤12mA; record REF_TRIP and CURRENT_SENSE.
• Supervised factory load test only: with power OFF, USB3V3→matrix J_POWER pin9 holds ALIVE high until CP13 watchdog exists. Explicit enable applies the bounded load; reset under overload must not release the latch. Remove the factory ALIVE link or hold it LOW to simulate heartbeat loss and prove OFF. Remove this link permanently at CP13.

Film the result

Show overload trip and latch setup with power OFF and display firmware revision.
Demonstrate: An overload shuts down and stays latched until a valid reset. Show the documented negative test and actual recorded readings.
Negative example: Apply the qualified overload: shutdown latches; clearing the fault does not re-enable.
Record actual readings, equipment, board/firmware revision and link positions.

CP13 — Heartbeat shutdown

Filmable win: Stopping the controller heartbeat switches the final rail OFF.

Installation

Add 14 components/posts, 28 local wires and 3 harness conductors. Earlier permanent work stays in place. Boards receiving parts: router-safety. • USB prototype source and prototype 3V3 links removed before final supplies connect. • Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Replace any supervised factory ALIVE link with the completed USB-powered hardware watchdog. Send heartbeat ≤100ms; stopping heartbeat or removing USB must switch rail OFF within 350ms and not restart automatically. Record scope timing.

Film the result

Show heartbeat shutdown setup with power OFF and display firmware revision.
Demonstrate: Stopping the controller heartbeat switches the final rail OFF. Show the documented negative test and actual recorded readings.
Negative example: Hold heartbeat static HIGH or LOW: watchdog must time out and disable power.
Record actual readings, equipment, board/firmware revision and link positions.

CP14 — Cutoff timing and transients

Filmable win: Measured cutoff timing and contact peaks meet the final limits.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

Firmware handoff

factory final-power qualification: explicit enable/reset; supervised ALIVE link only before CP13; no production authorization; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Scope current at a switched 100Ω load: source current cut off within 1ms, including frozen controller. Do not use UART timestamps as timing evidence.
• Independent OV disconnects: sweep 5V/3V3 buck interfaces with consumers isolated; verify actual thresholds and strict contact peak ≤5.25V during a bounded 12V failed-buck step. Test both guards separately before connecting any IC.

Film the result

Show cutoff timing and transients setup with power OFF and display firmware revision. Demonstrate: Measured cutoff timing and contact peaks meet the final limits. Show the documented negative test and actual recorded readings.
Negative example: Capture the documented cutoff transient; an excessive peak or slow shutdown is a failure.
Record actual readings, equipment, board/firmware revision and link positions.

CP15 — Bounded source and sink

Filmable win: Smart output drives valid levels and limits short-circuit current.

Installation

Add 124 components/posts, 185 local wires and 37 harness conductors. Earlier permanent work stays in place. Boards receiving parts: smart-core, smart-safety, measurement-a-buffers, bank-a-position-0. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

• Remove temporary ESP1 I2C/mux and USB-router-to-3V3_SYS links. ESP2 owns mux select and measurement; only final regulated rails supply bank electronics.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• P3 low ≤0.5V and high ≥3.5V with measured 2mA load. Short to GND and separately to 5.25V: ≤12mA; scope transient and regulator oscillation.

• Pulse 0.5/1/10Hz and duty 25/50/75%: frequency ±5%; duty ±5 percentage points; measure at the physical node.

Film the result

Show bounded source and sink setup with power OFF and display firmware revision. Demonstrate: Smart output drives valid levels and limits short-circuit current. Show the documented negative test and actual recorded readings.

Negative example: Use the qualified bounded short test: hardware clamps current and remains within limits.

Record actual readings, equipment, board/firmware revision and link positions.

CP16 — Pulse frequency

Filmable win: The real output changes its blink frequency.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Use a separately documented LED+1kΩ output fixture on P3. Do not connect the fixed battery to P3. OFF/reset/invalid state disables both output relays.

Film the result

Show pulse frequency setup with power OFF and display firmware revision.
Demonstrate: The real output changes its blink frequency. Show the documented negative test and actual recorded readings.
Negative example: Request a frequency outside 0.5–10Hz: reject configuration with output OFF.
Record actual readings, equipment, board/firmware revision and link positions.

CP17 — Pulse duty cycle

Filmable win: Changing duty cycle changes the real pulse waveform.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Scope 25%, 50%, 75% duty at a known frequency; error ≤5 percentage points. Frequency 0.5–10Hz ±5%; ≥5ms break-before-make; changing mode first turns output OFF.

Film the result

Show pulse duty cycle setup with power OFF and display firmware revision.
Demonstrate: Changing duty cycle changes the real pulse waveform. Show the documented negative test and actual recorded readings.
Negative example: Request duty outside 25–75%: reject configuration with output OFF.
Record actual readings, equipment, board/firmware revision and link positions.

CP18 — AND gate

Filmable win: The LED follows the AND truth table.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Test each selected gate truth table using protected input levels. Require 20ms stable input and ≤100ms output response. Confirm open inputs read LOW.

• Record all four AND input combinations using bounded removable links and DMM-confirmed input levels.

Film the result

Show and gate setup with power OFF and display firmware revision. Demonstrate: The LED follows the AND truth table. Show the documented negative test and actual recorded readings.

Negative example: AND inputs01,10 and00 must remain LOW; only11 may be HIGH. Record actual readings, equipment, board/firmware revision and link positions.

CP19 — OR gate

Filmable win: The LED follows the OR truth table.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Record all four OR input combinations: input ≤1V LOW, ≥3V HIGH; 20ms stability and output response ≤100ms.

Film the result

Show or gate setup with power OFF and display firmware revision. Demonstrate: The LED follows the OR truth table. Show the documented negative test and actual recorded readings.
Negative example: OR inputs00 must remain LOW;01,10 and11 must be HIGH. Record actual readings, equipment, board/firmware revision and link positions.

CP20 — NOT gate

Filmable win: The LED follows the NOT truth table.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Record LOW→HIGH and HIGH→LOW inversion; retain previous state in the input hysteresis band. Reset and invalid configurations leave output OFF.

Film the result

Show not gate setup with power OFF and display firmware revision. Demonstrate: The LED follows the NOT truth table. Show the documented negative test and actual recorded readings.

Negative example: NOT inputHIGH must produce LOW; inputLOW must produce HIGH.

Record actual readings, equipment, board/firmware revision and link positions.

CP21 — SR memory

Filmable win: Set persists, and reset wins.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Set output, remove the set input and verify retained state. Reset clears it. Simultaneous set/reset resolves to zero.

• Restart, rail OFF or configuration loss clears memory and leaves output disabled.

Film the result

Show sr memory setup with power OFF and display firmware revision. Demonstrate: Set persists, and reset wins. Show the documented negative test and actual recorded readings.

Negative example: Assert set and reset together: reset wins; rail-off loses the stored state.

Record actual readings, equipment, board/firmware revision and link positions.

CP22 — Current calibration

Filmable win: The ammeter matches known signed currents.

Installation

Add 10 components/posts, 18 local wires and 3 harness conductors. Earlier permanent work stays in place. Boards receiving parts: bank-a-position-0. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Current mode ±0/1/5/10mA: error ≤0.20mA+5%|reading|, ≥10 updates/s; voltage modes must leave ammeter switches open.
• INA226 with external 1Ω shunt 0x40 responds. Use a documented series test arrangement and removable leads; do not assume four-column capsule geometry supports every series arrangement.
• OFF, voltage or output mode opens both ammeter terminals. Verify leakage ≤10µA with test stimulus disconnected.

Film the result

Show current calibration setup with power OFF and display firmware revision.
Demonstrate: The ammeter matches known signed currents. Show the documented negative test and actual recorded readings.
Negative example: Use an incorrect current calibration: reject it rather than displaying an accepted value.
Record actual readings, equipment, board/firmware revision and link positions.

CP23 — Read loop current

Filmable win: Changing a real resistor changes measured series current.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• For smart-mode bench fixtures, disconnect both fixed battery leads and remove the initial kit loop. Output load: P3→1kΩ→LED anode, LED cathode→GND through removable leads. Logic inputs: HIGH=qualified3V3_SYS through1kΩ; LOW=GND through1kΩ. AND/OR exercise00,01,10,11; NOT exercise0,1. Never parallel another source with P3. Current tests use a documented series path, never an ammeter across the supply.

Firmware handoff

smart factory mode on ESP2: migrate OLED/ADC service to its isolated position bus; register outputs OFF until qualified; valid heartbeat required; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Power OFF to change the documented series fixture between 1kΩ and 470Ω. Compare OLED current with DMM; ≤0.20mA+5%|reading| error. Never put the ammeter directly across the supply.

Film the result

Show read loop current setup with power OFF and display firmware revision.
Demonstrate: Changing a real resistor changes measured series current. Show the documented negative test and actual recorded readings.
Negative example: Exit current mode: both ammeter terminals disconnect; verify rather than shorting the supply.
Record actual readings, equipment, board/firmware revision and link positions.

CP24 — Route one LED loop

Filmable win: The matrix completes one bounded LED loop.

Installation

Add 100 components/posts, 149 local wires and 36 harness conductors. Earlier permanent work stays in place. Boards receiving parts: matrix-switches, matrix-control. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• MT supply 10.0–12.6V, digital high ≥3.3V, low ≤0.8V; reset and CS active high.

• Sweep all X addresses and Y buses, proving the non-monotonic X address map. Banks share BUS2…BUS7.

• Boot/reset/watchdog expiry: all crosspoints open. Disable rail, clear, configure, precheck, then enable; never configure live.

• Scope full clear + 22-node configuration ≤50ms at the physical strobe pins; verify Ron at 0/2.5/5V.

• Calibrate with 1kΩ 0.1%; distinguish open, short, 220Ω, 1kΩ and diode polarity; resistance error ≤10% after calibration.

Film the result

Show route one led loop setup with power OFF and display firmware revision.
Demonstrate: The matrix completes one bounded LED loop. Show the documented negative test and actual recorded readings.
Negative example: Remove enable or a required acknowledgement: matrix LED loop remains OFF.
Record actual readings, equipment, board/firmware revision and link positions.

CP25 — Inspect a routed jumper

Filmable win: The matrix test tools distinguish a routed short from an open.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Discharge 1000µF initially at 5V to <0.1V in 3s; keep total analog pin current ≤12mA.
• Any short, unknown/invalid configuration or stale routing generation prevents rail enable; the precheck itself leaves rail off.
• With all power off remove only the fixed battery patch leads; retain every soldered wire. Connect A1 NODE ribbon. In matrix mode no fixed source or fixture stimulus may remain connected.

Film the result

Show inspect a routed jumper setup with power OFF and display firmware revision.
Demonstrate: The matrix test tools distinguish a routed short from an open. Show the documented negative test and actual recorded readings.
Negative example: Select a known open diagnostic path: return rejection rather than a false continuity pass.
Record actual readings, equipment, board/firmware revision and link positions.

CP26 — Reject an invalid route

Filmable win: Invalid or stale routing requests keep power OFF.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Show a valid second battery position and corresponding kit arrangement. Invalid topology or stale routing generation cannot enable power.

Film the result

Show reject an invalid route setup with power OFF and display firmware revision.

Demonstrate: Invalid or stale routing requests keep power OFF. Show the documented negative test and actual recorded readings.

Negative example: Submit the documented invalid topology: reject it without energizing.

Record actual readings, equipment, board/firmware revision and link positions.

CP27 — Qualify A2

Filmable win: The same capsule works at A2 without changes.

Installation

Add 68 components/posts, 126 local wires and 28 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a2, measurement-a-buffers, bank-a-position-1. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

• Before powering U_A_BUFFERS, fit removable GND ties at noninverting inputs pin10/pin12 for the future third position. Remove these ties with power OFF at the next expansion checkpoint before attaching its divider inputs.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Repeat CP01 electrical mapping for A2/A3; test NODE00–NODE10 and positions 0/1/2.

• Three active capsules meet meter update rates and output timing concurrently; fourth active request refused.

• Repeat screen, voltage, pulse, gate, memory, current and leakage tests independently on each new card. Three active capsules meet all update/timing limits.

Film the result

Show qualify a2 setup with power OFF and display firmware revision. Demonstrate: The same capsule works at A2 without changes. Show the documented negative test and actual recorded readings.

Negative example: Move or remove the capsule from A2: power shuts down before reconfiguration.

Record actual readings, equipment, board/firmware revision and link positions.

CP28 — Qualify A3

Filmable win: Three bank-A positions operate with no address collisions.

Installation

Add 61 components/posts, 107 local wires and 26 harness conductors. Earlier permanent work stays in place. Boards receiving parts: a3, measurement-a-buffers, bank-a-position-2. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Repeat the single-position acceptance at A3; exercise all three bank-A capsules concurrently. Every buffer B-side remains local, and every position retains the same mode capabilities.

Film the result

Show qualify a3 setup with power OFF and display firmware revision. Demonstrate: Three bank-A positions operate with no address collisions. Show the documented negative test and actual recorded readings.

Negative example: Move or remove the capsule from A3: power shuts down before reconfiguration.

Record actual readings, equipment, board/firmware revision and link positions.

CP29 — Qualify B1

Filmable win: A capsule moves to B1 without rotation or polarity changes.

Installation

Add 111 components/posts, 190 local wires and 43 harness conductors. Earlier permanent work stays in place. Boards receiving parts: b1, measurement-b-precision, measurement-b-buffers, bank-b-position-3. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Repeat module mapping for B1/B2/B3; verify mirrored service order without renumbering holes.

• Test NODE00–NODE21 and all six smart positions, any combination of three active; no unintended bank-to-bank connection.

Film the result

Show qualify b1 setup with power OFF and display firmware revision. Demonstrate: A capsule moves to B1 without rotation or polarity changes. Show the documented negative test and actual recorded readings.

Negative example: Move the unchanged-orientation capsule to B1: old-position authorization must not energize it.

Record actual readings, equipment, board/firmware revision and link positions.

CP30 — Qualify B2

Filmable win: The same capsule passes acceptance at B2.

Installation

Add 68 components/posts, 126 local wires and 28 harness conductors. Earlier permanent work stays in place. Boards receiving parts: b2, measurement-b-buffers, bank-b-position-4. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

• Before powering U_B_BUFFERS, fit removable GND ties at noninverting inputs pin10/pin12 for the future third position. Remove these ties with power OFF at the next expansion checkpoint before attaching its divider inputs.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Repeat single-position voltage, output, current, OLED and movement shutdown checks at B2 with unchanged capsule orientation.

Film the result

Show qualify b2 setup with power OFF and display firmware revision. Demonstrate: The same capsule passes acceptance at B2. Show the documented negative test and actual recorded readings.

Negative example: Remove the capsule from B2: stale identity cannot keep output enabled.

Record actual readings, equipment, board/firmware revision and link positions.

CP31 — Qualify B3

Filmable win: All six positions work while only three capsules are active.

Installation

Add 61 components/posts, 107 local wires and 26 harness conductors. Earlier permanent work stays in place. Boards receiving parts: b3, measurement-b-buffers, bank-b-position-5. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

router+smart production interlock: final power qualified; valid topology; bounded matrix tool current; fresh ACKs; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Moving a capsule drops rail before configuration; position identity is independent of capsule identity.

• Repeat single-position checks at B3; attempt a fourth active capsule and confirm rejection with power OFF.

Film the result

Show qualify b3 setup with power OFF and display firmware revision. Demonstrate: All six positions work while only three capsules are active. Show the documented negative test and actual recorded readings.

Negative example: Request a fourth active capsule: reject it and retain OFF.

Record actual readings, equipment, board/firmware revision and link positions.

CP32 — Pi communication

Filmable win: Pi commands reach the controllers with revisioned acknowledgements.

Installation

Add 15 components/posts, 30 local wires and 1 harness conductors. Earlier permanent work stays in place. Boards receiving parts: integration, router-core, smart-core. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

complete Pi/router/smart session and NFC interlock; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Pi separately powered: its 5V must not connect to bench 5V. Test all four USB/bench-power combinations for backfeeding.

• Verify physical Pi pin mapping and button debounce. Readiness alone does not energize; explicit request is required. Back/reset/restart returns OFF.

Film the result

Show pi communication setup with power OFF and display firmware revision.
Demonstrate: Pi commands reach the controllers with revisioned acknowledgements. Show the documented negative test and actual recorded readings.
Negative example: Use a mismatched, expired or duplicate acknowledgement: communication cannot authorize enable.
Record actual readings, equipment, board/firmware revision and link positions.

CP33 — Session-gated enable

Filmable win: Only a valid session and fresh acknowledgements permit enable.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

complete Pi/router/smart session and NFC interlock; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Factory Pi demonstration uses explicit supervised approval; production mode remains blocked until NFC/full integration acceptance.

Film the result

Show session-gated enable setup with power OFF and display firmware revision.
Demonstrate: Only a valid session and fresh acknowledgements permit enable.
Show the documented negative test and actual recorded readings.
Negative example: Expire the session or lose communication: output shuts down without autonomous restoration.
Record actual readings, equipment, board/firmware revision and link positions.

CP34 — Known NFC identity

Filmable win: A known NFC identity participates in approval.

Installation

Add 1 components/posts, 7 local wires and 0 harness conductors. Earlier permanent work stays in place. Boards receiving parts: integration. • USB prototype source and prototype 3V3 links removed before final supplies connect.

• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.

• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

complete Pi/router/smart session and NFC interlock; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• RC522 SPI mode 0, MSB-first, ≤1MHz initial clock: known/unknown/absent tag correctly reported; NFC failure leaves session unapproved.

• Known identity, valid lesson and fresh controller ACKs permit a separate explicit enable request. NFC alone never energizes.

Film the result

Show known nfc identity setup with power OFF and display firmware revision.

Demonstrate: A known NFC identity participates in approval. Show the documented negative test and actual recorded readings.

Negative example: Present a known NFC identity without a valid session: NFC alone cannot energize.

Record actual readings, equipment, board/firmware revision and link positions.

CP35 — Reject NFC failures

Filmable win: Unknown, absent or failed NFC cannot authorize power.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

complete Pi/router/smart session and NFC interlock; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Known tag allows the subsequent readiness checks; unknown/absent/error clears approval. Explicit Energizar is still required after a successful tag.

Film the result

Show reject nfc failures setup with power OFF and display firmware revision.
Demonstrate: Unknown, absent or failed NFC cannot authorize power. Show the documented negative test and actual recorded readings.
Negative example: Present unknown, absent or failed NFC: authorization is denied and power stays OFF.
Record actual readings, equipment, board/firmware revision and link positions.

CP36 — Complete lesson

Filmable win: An enclosed full lesson completes with its fault cases demonstrated.

Installation

Add 0 components/posts, 0 local wires and 0 harness conductors. Earlier permanent work stays in place. • USB prototype source and prototype 3V3 links removed before final supplies connect.
• Use checkpoint-specific bounded test loads; stop and isolate power before changing any link.
• Remove all fixed source patch leads before matrix routing. Source node selection now belongs to the qualified matrix topology.

Firmware handoff

complete Pi/router/smart session and NFC interlock; revision5 bank acquisition, carrier interfaces and selector ownership supersede legacy ADC pin mapping

Measured acceptance

• Graph, recognition, predicted current, routing and all active capsule acknowledgements must pass before enable.
• USB unplug/reset or missed heartbeat disables rail and smart outputs within 1s; every restart begins OFF.
• 30-minute repeated lessons with no reset, incorrect route or false trip. Series-circuit tests use explicitly documented removable leads where geometry requires them.
• Repeat hardware cutoff ≤1ms, strict transient contact peaks ≤5.25V, 30-minute operation, no backfeed, leakage ≤10µA and every six-position mode. Record oscilloscope readings as text; screenshots remain external.
• Verify enclosure inaccessibility, cable strain relief and all purchased-part fit gates. Production release remains pending until actual evidence is recorded.

Film the result

Show complete lesson setup with power OFF and display firmware revision.
Demonstrate: An enclosed full lesson completes with its fault cases demonstrated.
Show the documented negative test and actual recorded readings.
Negative example: Run the complete lesson fault cases, including communication loss and invalid topology: each disables power.
Record actual readings, equipment, board/firmware revision and link positions.